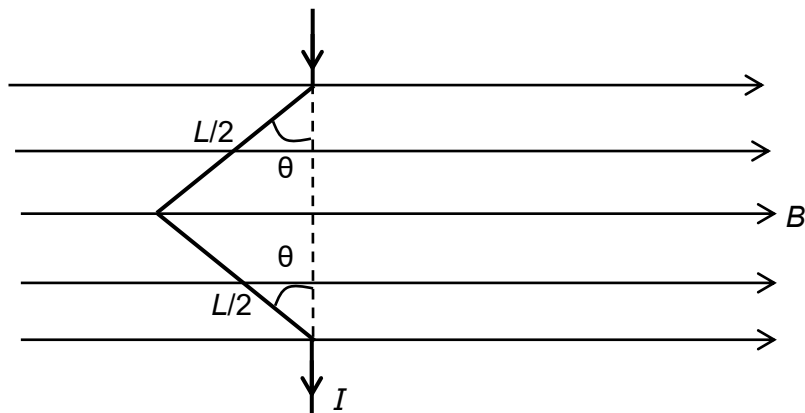


- 25 The diagram shows a V-shaped wire in the magnetic field of flux density B . The length of the wire in the field is L and the first half of the wire is inclined with an angle θ normal to the field direction. The current in the wire is I .



Which row gives the magnitude and direction of the force acting on the wire?

	magnitude	direction
A	$BIL\cos \theta$	into of the page
B	$BIL\cos \theta$	out of the page
C	$BIL\sin \theta$	into the page
D	$BIL\sin \theta$	out of the page

Ans: B

Resolve the wire to be perpendicular to the B-field, $F = BI(L/2 \cos\theta + L/2 \cos\theta) = BIL\cos\theta$

Using FLHR, direction is out of the page.

- 26 A 120 turn coil of area $5.0 \times 10^{-4} \text{ m}^2$ is placed in a magnetic field of strength $5.0 \times 10^{-2} \text{ T}$. The